

# Subhransu S. Bhattacharjee

School of Computing, The Australian National University, Australia

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PhD candidate in Artificial Intelligence at ANU specialising in world models, generative models, and embodied AI for autonomous systems.  
Experience in uncertainty quantification, spatial reasoning, agentic systems, computer vision, and natural language processing.

## Education

### Doctor of Philosophy

School of Computing, Australian National University, Australia

- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*
- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

### Artificial Intelligence

Apr 2023 – Present

### Bachelor of Engineering

School of Engineering, Australian National University, Australia

- **Major:** Mechatronic Systems Engineering, *First Class Honours*
- **Minors:** Mathematics and Electronic Communication Systems; summer school at the London School of Economics
- **Certifications:** [Game Theory, Stanford](#); [Machine Learning Production, DeepLearning.AI](#); [Project Management, Google](#)
- **Thesis project:** [Whiplash Gradient Descent Dynamics](#)

### Mechatronics

Jul 2018 – Dec 2022

## Internship Experience

### Microsoft Corporation

Applied Scientist PhD Research Intern, Natural Language Processing

- Collaborated across Microsoft AI and the India Development Center on a post-training RAG module for Copilot and Foundry, integrating agentic tool selection and multi-chat history retrieval for grounded generation.

May 2026 – July 2026

### Optiver APAC

Quantitative Research Intern, Machine Learning

- Built a real-time ML research pipeline for Korean market microstructure, including feature generation, monitoring, and walk-forward backtesting for rapid strategy evaluation.

Nov 2024 – Feb 2025

### JP Morgan Chase & Co.

Quantitative Research Intern, Technology and Risk

- Built a transformer-based question-answering pipeline over large financial document collections, including document indexing and retrieval components for client-facing research workflows.

Dec 2021 – Feb 2022

### Defence Research & Development Organisation, India

Summer Research Trainee, Radar Signal Processing

- Implemented an FPGA-based multiprocessor interface for real-time data exchange and long-range passive radar signal analysis using extended Kalman matched filtering.

Apr 2020 – Jul 2020

## Academic Experience

### Student Hub, Australian National University

Quantitative Researcher for the ANU Student Hub

- Developed an AI-assisted workflow for timetable optimisation and Student Hub usability improvement.

Feb 2026 – Apr 2026

### Research School of Management, Australian National University

Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan

- Built assessment frameworks for banks' AI adoption and organisational responses to technological change in an industry-linked project with Commonwealth Bank of Australia.
- Analysed open-banking regimes using a dynamic-capabilities lens to inform strategy recommendations.

Sep 2023 – Sep 2024

### CIICADA Lab, Australian National University

Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen

- Studied feedback-based optimisation methods for systems without closed-form solutions, contributing to publications at ASCC 2022 and ANZCC 2021.

Dec 2022 – Feb 2023

### School of Computing, Australian National University

Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley

- Investigated invertibility of differentiable neural mappings using neural networks; achieved a 72% hit rate under an RMSE criterion using dense positional encoding.

Mar 2022 – Jun 2022

## Selected Publications

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**Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome:** *Believing is Seeing: Unobserved Object Detection using Generative Models*. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [Project page](#).  
*Uses pre-trained generative models to infer spatio-semantic distributions of occluded and out-of-frame objects across 2D, 2.5D, and 3D scenes.*

**Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome:** *FlatLands: Generative Floorplan Completion From a Single Egocentric View*. [Paper](#).

*Introduces a benchmark for single-view bird's-eye-view floor completion and evaluates uncertainty-aware generative mapping from partial egocentric observations.*

**Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome:** *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models*. *arXiv:2510.11014*, 2025. Under review. [Preprint](#).

*Samples environment priors from partial observations using pretrained generative models.*

*Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.*

**Subhransu S. Bhattacharjee & Ian Petersen:** *Analysis of the Whiplash Gradient Descent Dynamics*. *Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: [10.1002/asjc.3153](#).

*A feedback-based inertial gradient method with adaptive damping achieves provable polynomial and exponential convergence rates that extend classical optimisation theory to high-dimensional settings.*

## Achievements

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**2026** Google Research, Granted with ANU Robotics \$76,000 for the Development of World Models for Task and Motion Planning

**2025** Invited Research Talk at the Dyson Robotics Laboratory, Imperial College London

**2025** Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.

**2025** Invited attendee, Citadel & Citadel Securities PhD Summit, London.

**2025** Invited attendee, Citadel Securities Australia, Summer Networking Event, Sydney.

**2024** Optiver PhD Quant Lab Program (selected participant).

**2023** ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.

**2022** Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.

**2022** Course highest in ENGN4627: Robotics.

**2021** High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.

**2020** ANU Chancellor's International Scholarship for academic merit for transfer students.

**2019** Chancellor's Achiever and Special Achiever Award for Academic Merit, Vellore Institute of Technology, India

## Teaching & Service

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### Teaching

- **Senior Tutor, Introduction to Machine Learning; Professional Practices: Responsible Leadership** — School of Computing, ANU (2025 – 2026): Taught machine-learning mathematics and optimisation; ran tutorials, marked exams, and assessed projects.
- **Head Tutor, Building Cyber-Physical Systems** — School of Cybernetics, ANU (Mar 2025 – Nov 2025): Co-designed and delivered project-based modules in microprocessors, robotics, and machine learning; led NAO robot labs; coordinated assessments.
- **Tutor, Power Systems & Control Theory** — School of Engineering, ANU (Jul 2022 – Sep 2023): Ran labs for ENGN8824, workshops for ENGN4628, and targeted tutoring for ENGN4625.

### Service

**Reviewing:** ECCV 2026; NeurIPS 2026; CVPR 2026; ICRA 2025–2026; AAAI 2025; IROS 2025; Asian Journal of Control (2023); American Control Conference (2022); Australia & New Zealand Control Conference (2021–2022).

**Volunteering:** Friends of Tribal Society volunteer (2017); Set4ANU HDR mentor (2024); TechLauncher manager (2024).

## Technical Skills

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**Programming:** Python; SQL; C; MATLAB; Bash; Embedded C;  $\LaTeX$ .

**Computer Vision & Robotics:** OpenCV; Open3D; PyTorch3D; TorchVision; ROS2; COLMAP; Trimesh; detection; segmentation; monocular depth estimation; 3D reconstruction; robotic perception; spatio-semantic mapping.

**Systems & Infrastructure:** Linux; Git; Docker; Apptainer; SLURM; CUDA; Numba; CuPy; `torch.compile`; ONNX Runtime; Weights & Biases; AWS — EC2, S3.

**Training, Evaluation & Deployment:** mixed-precision training; profiling; benchmarking; uncertainty estimation; ablation studies; distillation; reproducible experimentation; AMD Vivado for edge deployment and signal processing.

**Data & ML Infrastructure:** NumPy; pandas; SciPy; scikit-learn; PySpark; REST APIs; backtesting frameworks.

**Large Language Models:** transformer models; supervised fine-tuning; RLHF; RLAIF; DPO; RAG; vector databases.

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